

KARAN CHAWLA DORA

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EDUCATION

University of Toronto

Bachelor of Applied Science — Engineering Science (cGPA: **3.95/4.0**)

Toronto, ON

2025 – 2030

Interests: autonomous driving, state estimation, motion planning, control systems

EXPERIENCE

aUToronto — Toronto Autonomous Vehicle Team [↗](#)

Toronto, ON

3D Mapping Lead (promoted)

Jul. 2026 – Present

- Produced high-definition maps of UTIAS by leading a **2-person team** building LiDAR SLAM pipelines with LIO-SAM and integrating GPS and inertial measurement unit (IMU) data.
- Extend autonomous-vehicle mapping to public-road driving logs, developing dynamic-object removal workflows to filter moving objects from LiDAR scans and preserve static landmarks for localization.

State Estimation Developer

Sep. 2025 – Jul. 2026

- Enabled lane localization during GPS dropouts by developing a C++ pipeline with an Iterated Extended Kalman Filter (IEKF), processing **100+ lane observations per 5 m** to estimate vehicle position.
- Helped the team earn **1st place in localization and overall** at SAE AutoDrive Challenge R2Y5 by validating LiDAR and lane localization against ground-truth GPS using Python, ROS 2 bag analysis, and CI workflows.

FIRST Robotics Competition [↗](#)

Ottawa, ON

Robotics Team Captain & Software Development Lead

Jul. 2021 – Jun. 2025

- Helped raise average autonomous scoring by approximately **3x** from the first competition to provincials by building C++ routines combining AprilTag computer vision, wheel odometry, and trajectory generation.
- Improved robot mechanism precision by tuning PID feedback and feedforward controllers for autonomous and driver-controlled operation, integrating control software with competition robot hardware.
- Helped earn **World Championship qualification** and **2 Provincial Finalist finishes** by leading a **90+ member team** through robot design, manufacturing, and software development.

RESEARCH EXPERIENCE

Autonomous Space Robotics Lab (ASRL) [↗](#)

Toronto, ON

Robotics Researcher — UTIAS — Supervisor: Tim Barfoot

May 2026 – Present

- Enabled **400+ m line-of-sight connectivity**, overcoming a prior **5 m robot-separation limit**, by replacing centralized Wi-Fi with an OpenWrt mesh network across **4 robots and 1 base station**.
- Validated multi-robot network reliability through field tests of non-line-of-sight communication and recovery from disconnected nodes, verifying operation beyond centralized Wi-Fi coverage.
- Enabled monitoring for planned multi-robot dataset collection by building a ROS 2 node that publishes network telemetry and battery health, providing visibility into robot communication and power status.

Bruce Lab — University of Ottawa [↗](#)

Ottawa, ON

Research Assistant

Aug. 2024

- Reduced system response time by approximately **80%** and steady-state error by approximately **40%** versus the baseline by implementing C++ PID controllers for fuel-cell electrodeposition experiments.
- Cut experiment completion time by approximately **33%** by building a Python GUI to automate equipment operation, sensor logging, and data preparation, replacing manual steps in the research workflow.
- Expanded the operating voltage range by approximately **5x** by refining an Arduino-based electrochemical measurement circuit and its control workflow for fuel-cell research experiments.

PROJECTS

Q-Learning Trainer [↗](#) | Python, FastAPI, NumPy, JavaScript

Jun. 2026

- Enabled reinforcement learning experiments for grid navigation by building a Python/FastAPI Q-learning simulator with custom maps, adjustable training parameters, and live WebSocket visualization.
- Delivered a publicly accessible training dashboard by deploying with Docker on Fly.io, isolating user sessions, and adding pytest regression tests for map loading, path reachability, and model evaluation.

TECHNICAL SKILLS

Languages: C++, Python, C, Bash, Java, JavaScript, TypeScript

Autonomy & Control: State estimation, sensor fusion, SLAM, IEKF, computer vision, trajectory generation, PID control

Robotics & Tools: ROS 2, LiDAR, GPS/IMU, Linux, Git, CI testing, Docker, NumPy, pandas, pytest, Arduino